

17/18 Sem 2 Q2(c)

$$\text{Given } \underline{u} \times (\underline{v} \times \underline{w}) = (\underline{u} \cdot \underline{w}) \underline{v} - (\underline{u} \cdot \underline{v}) \underline{w}$$

$$\text{Find } \underline{z} \times (\underline{v} \times \underline{w}) \text{ if } \underline{z} = \frac{\underline{u} \times \underline{v}}{\underline{u} \cdot (\underline{v} \times \underline{w})} \text{ and } \underline{u} \cdot \underline{v} \times \underline{w} \neq 0.$$

$$\underline{z} \times (\underline{v} \times \underline{w}) = (\underline{z} \cdot \underline{w}) \underline{v} - (\underline{z} \cdot \underline{v}) \underline{w}$$

$$= \left(\left(\frac{\underline{u} \times \underline{v}}{\underline{u} \cdot (\underline{v} \times \underline{w})} \right) \cdot \underline{w} \right) \underline{v} - \left(\left(\frac{\underline{u} \times \underline{v}}{\underline{u} \cdot (\underline{v} \times \underline{w})} \right) \cdot \underline{v} \right) \underline{w}$$

$$= \left(\frac{1}{\underline{u} \cdot (\underline{v} \times \underline{w})} (\underline{u} \times \underline{v} \cdot \underline{w}) \right) \underline{v} - \left(\frac{1}{\underline{u} \cdot (\underline{v} \times \underline{w})} (\underline{u} \times \underline{v} \cdot \underline{v}) \right) \underline{w}$$

$$= \left(\frac{1}{\underline{u} \cdot (\underline{v} \times \underline{w})} (\underline{v} \times \underline{w} \cdot \underline{u}) \right) \underline{v}$$

$$= \left(\frac{1}{\underline{u} \cdot \underline{v} \times \underline{w}} (\underline{u} \cdot \underline{v} \times \underline{w}) \right) \underline{v}$$

$$= 1 \cdot \underline{v} = \underline{v}$$